
Measurements in Optical Fiber Communications

Experimental Aspects

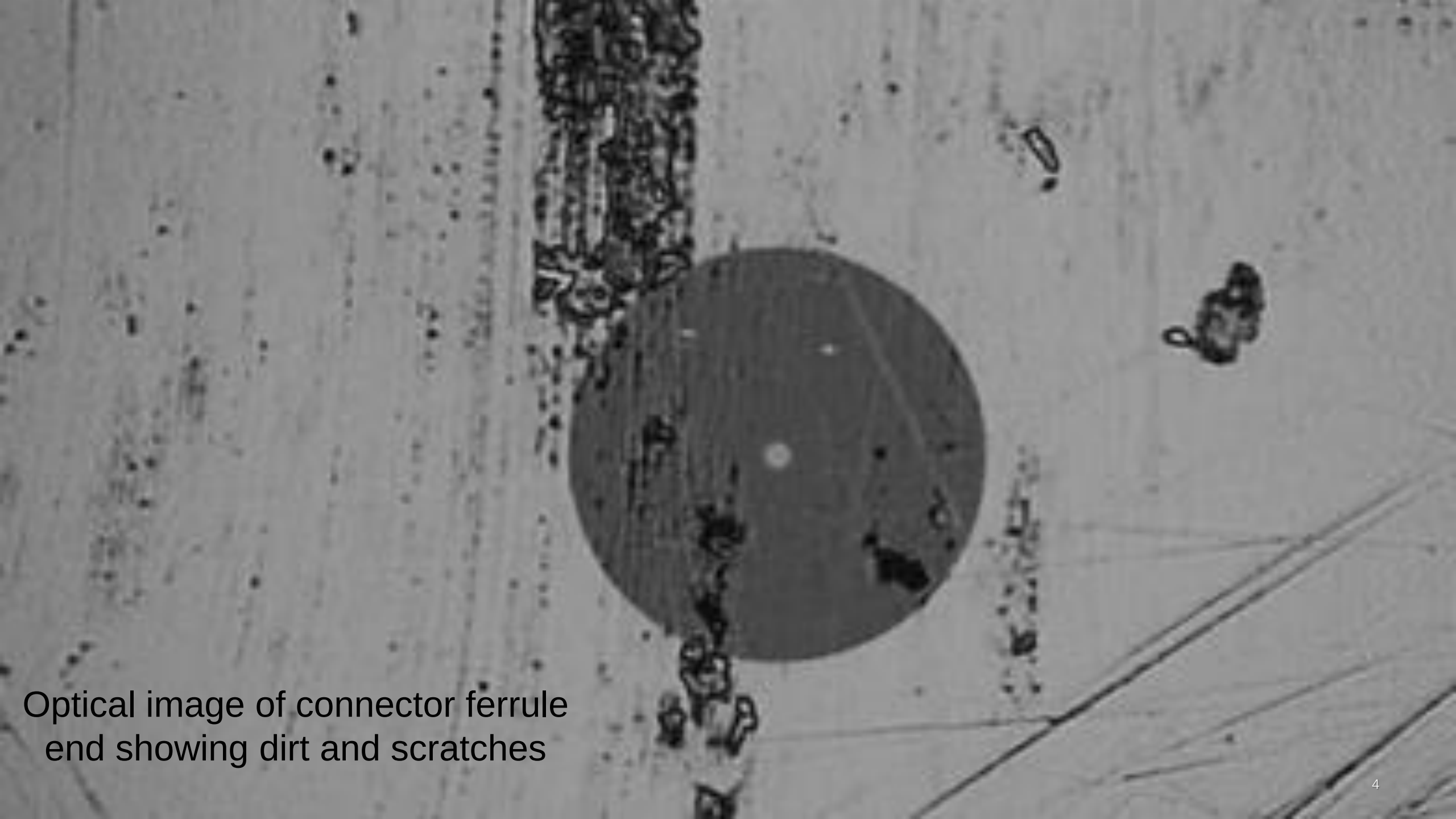
Optical Power



Pigtail and Connectorized Devices

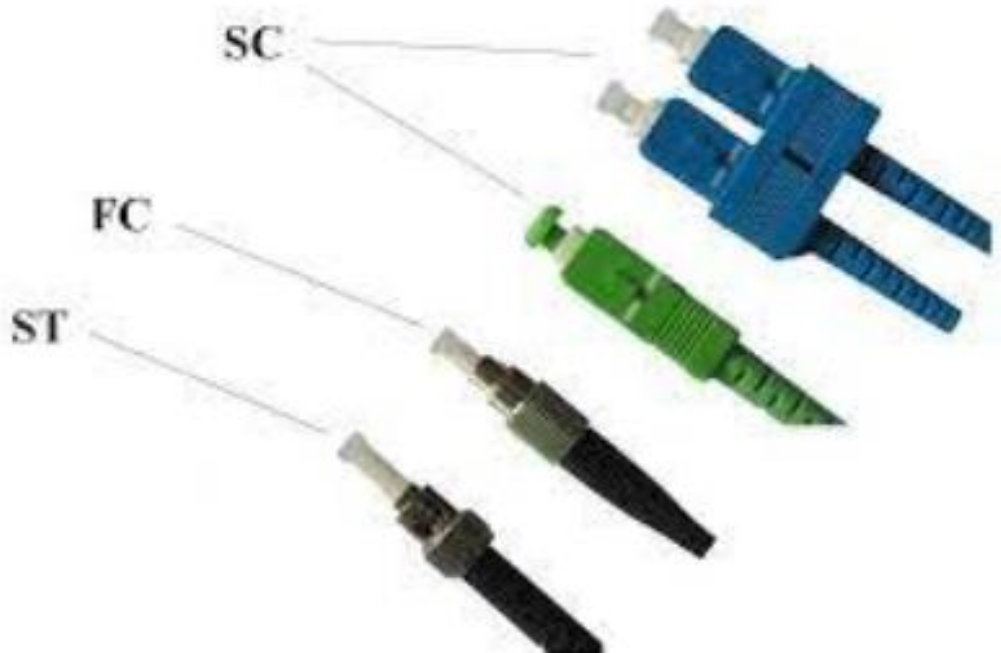


Video Inspection Probe



Optical image of connector ferrule
end showing dirt and scratches

Fiber Optic Assemblies



LC Connector



SC Connector



ST Connector



MTRJ



MU



E 2000 Connector

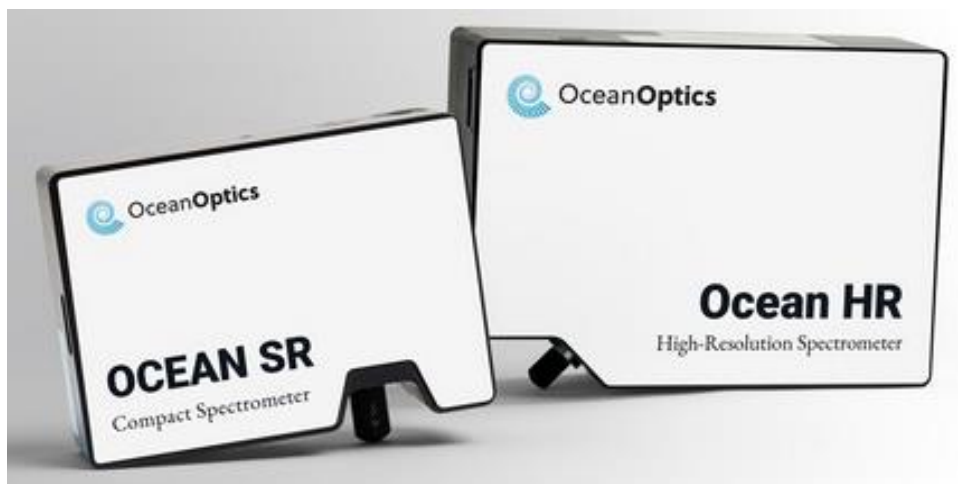


SC Fiber Optic Adapter



Optical Spectrum Analyzer

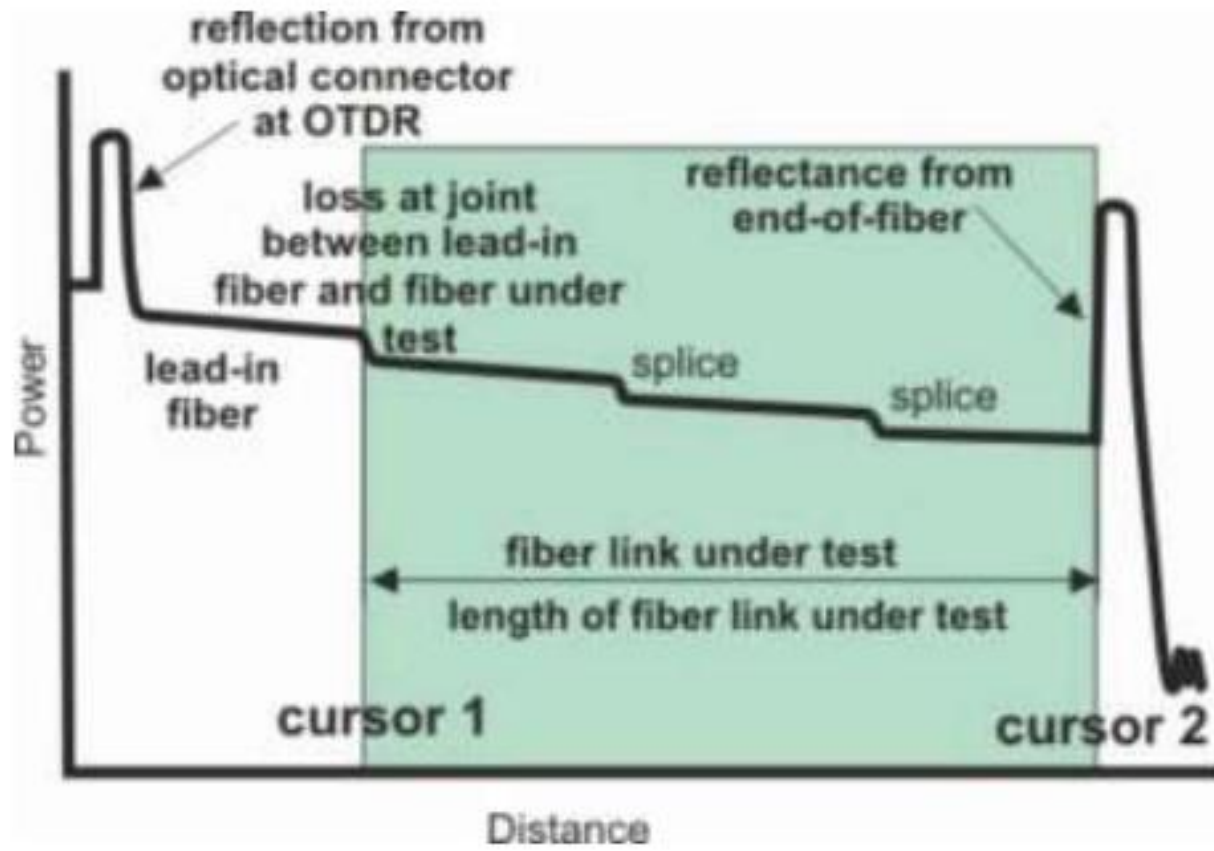




OSA – Visible Light and Near InfraRed (NIR)

Specification	SR2	SR4	SR6
Wavelength Range (configuration-dependent) <u>See Manual for full specifications</u>	220 - 1050 nm	220 - 1050 nm	185 - 1100 nm
Resolution (FWHM) (configuration-dependent) <u>See Manual for full specifications</u>	0.28 - 11.52 nm	0.42 - 11.34 nm	0.36 - 11.52 nm
Integration Time	1 μ s - 2 s	3.8 ms - 10 s	7.2 ms - 5 s

OTDR



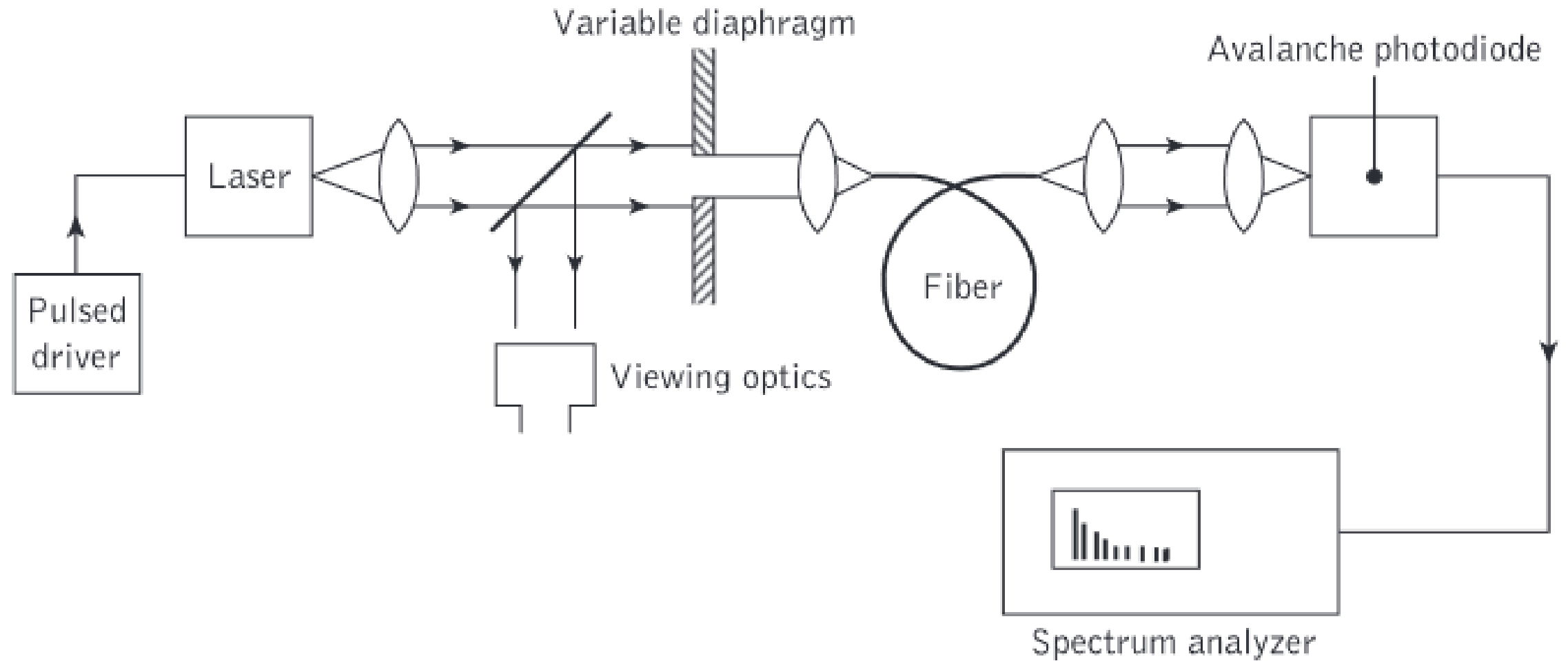


Figure 14.10 Experimental setup for making fiber dispersion measurements in the frequency domain using a pulsed laser source

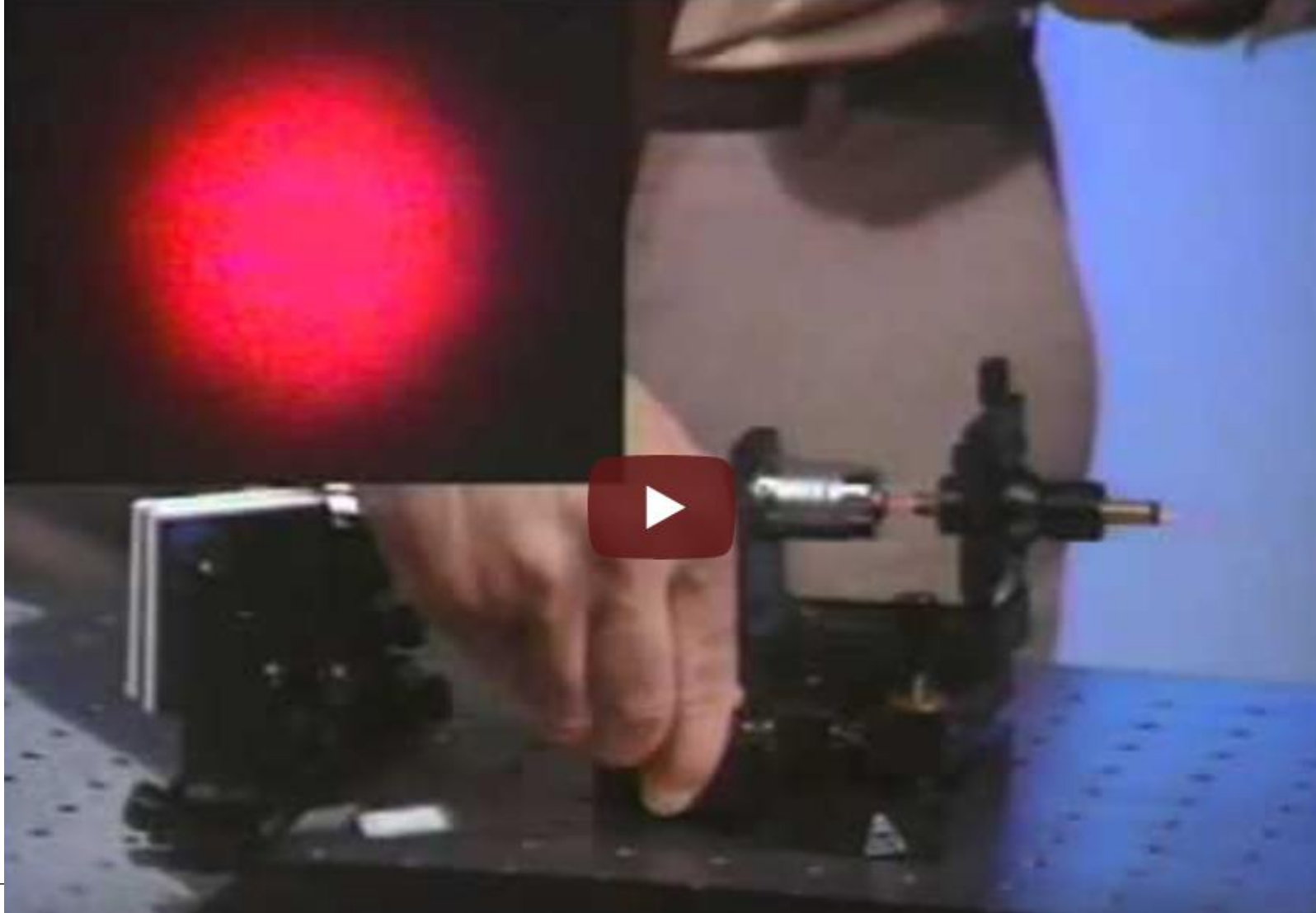
WDM Source



**PRO8000 Chassis
with 8 Modules**

Fiber Mode Propagation

<https://www.youtube.com/watch?v=45X0puB3YK0>



Optics: Single mode fiber | MIT Video Demonstrations in Lasers and

Fiber Mode Propagation

<https://www.youtube.com/watch?v=uzXLhTW9wWQ>



Optics: Multi-mode fiber | MIT Video Demonstrations in Lasers and Optics